

Airframe Technology Master

Checkstress

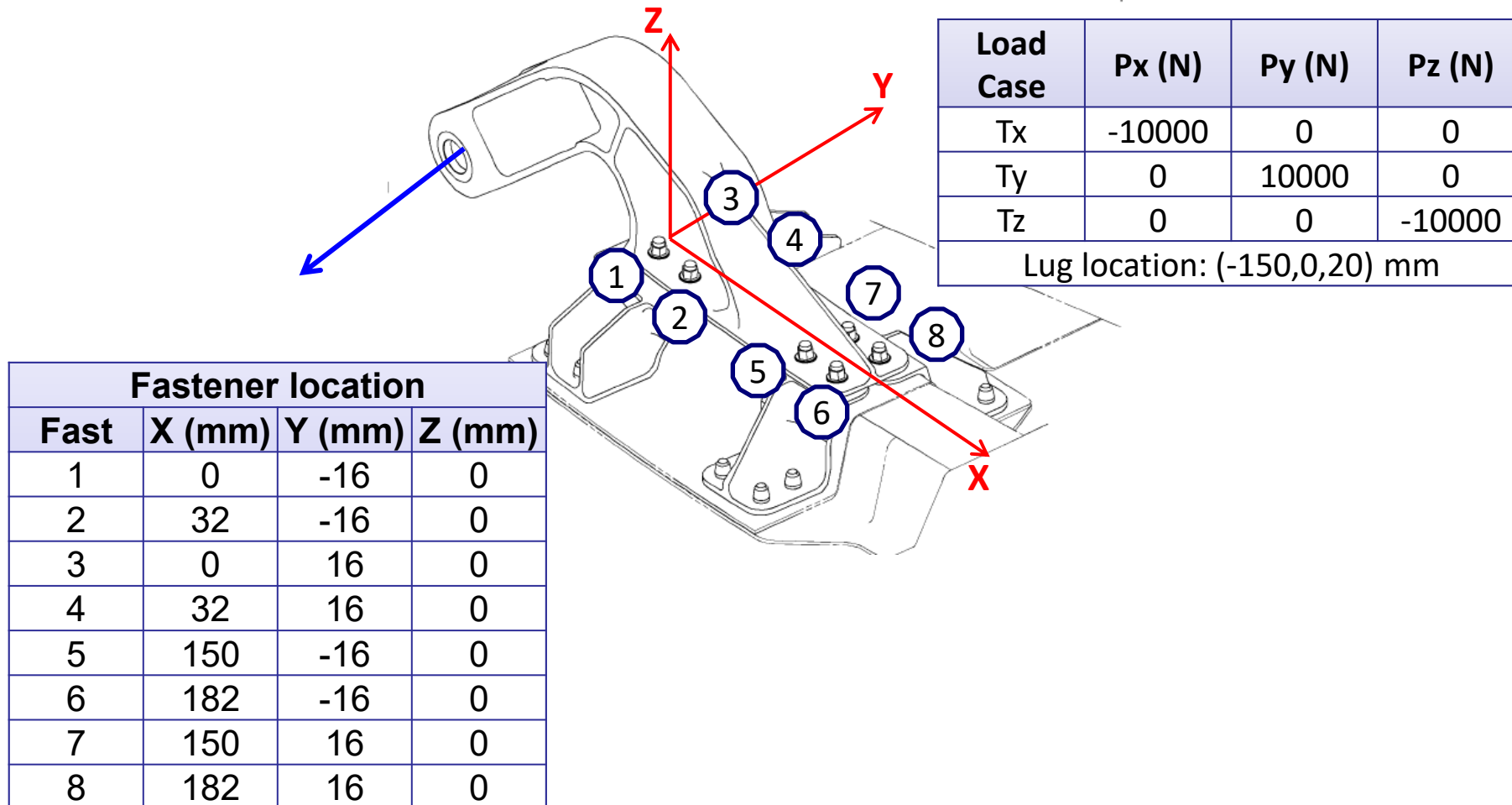
Lecture 13. Metallic Sizing-Exercise (Bolt Group)

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11 June 2025

Bolt Group Example

Obtain the fasteners load of the following fitting: (**Exercise 3**)



All fasteners are equal: same diameter, same plate thickness

Without formulas: (1st SOLUTION, CONCEPTUAL)

- **For Tx load case:**
 - Which is the most loaded fastener in tension? Why?
 - Which part of the fitting is in compression? Why?
 - Are all loaded with equal in-plane load? Why?

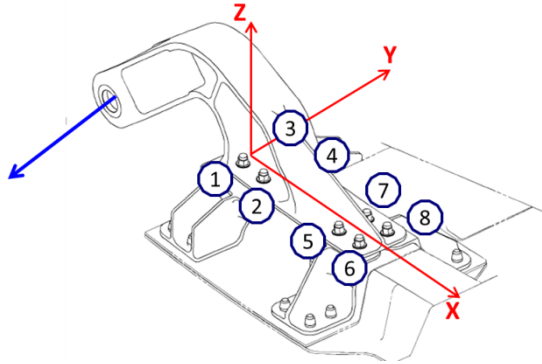
- **For Ty load case:**
 - Which is the most loaded fastener in tension? why?
 - Which part of the fitting is in compression? Why?
 - Are all loaded with equal in-plane load? Why?

- **For Tz load case:**
 - Which is the most loaded fastener in tension? why?
 - Which part of the fitting is in compression? Why?
 - Are all loaded with equal in-plane load? Why?

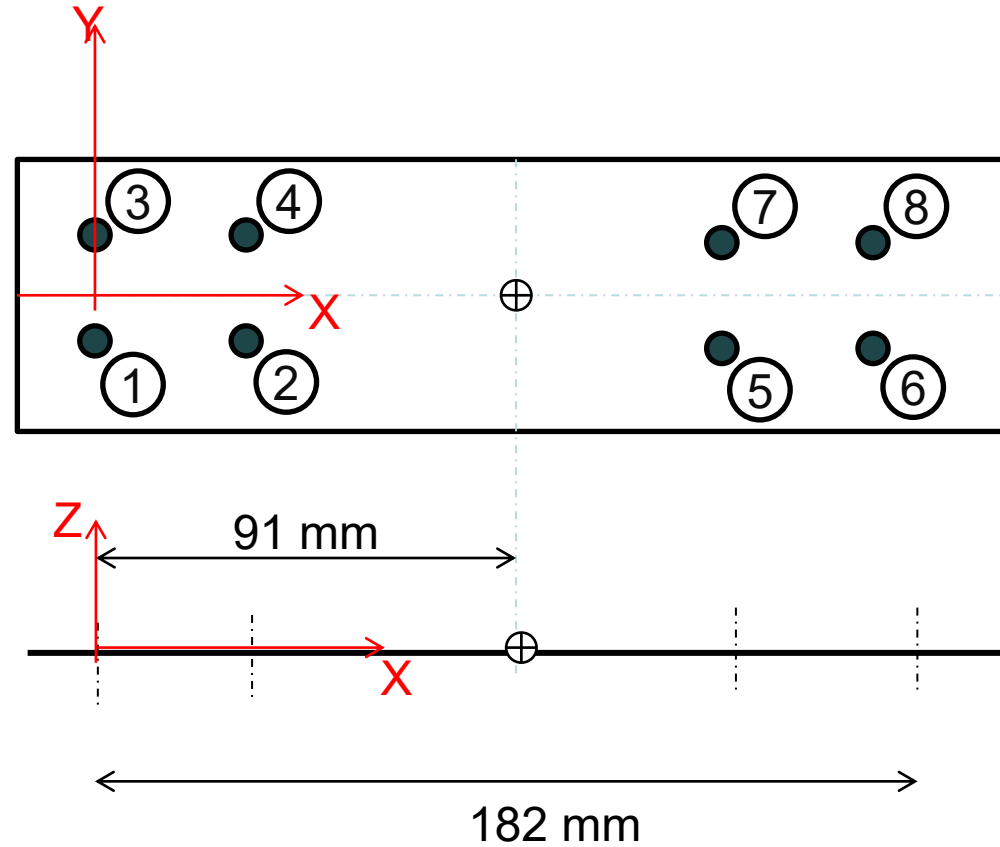
- Which load case is most critical in tension?
- Which one for in-plane loads?

Bolt Group Example

Find the centroid (2ND SOLUTION: PREDESIGN)



Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

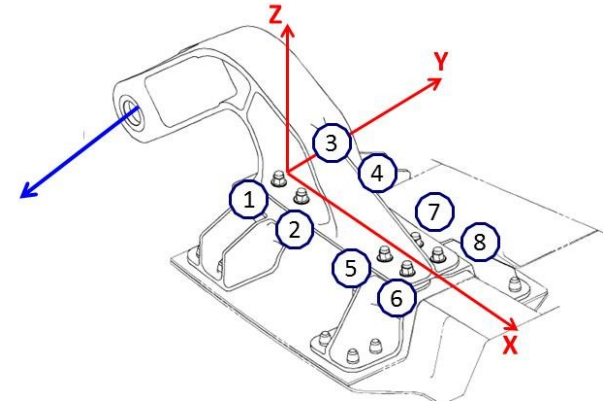


All fasteners are equal: same diameter, same plate thickness

Bolt Group Example

1. Bolt group centroid:

Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0



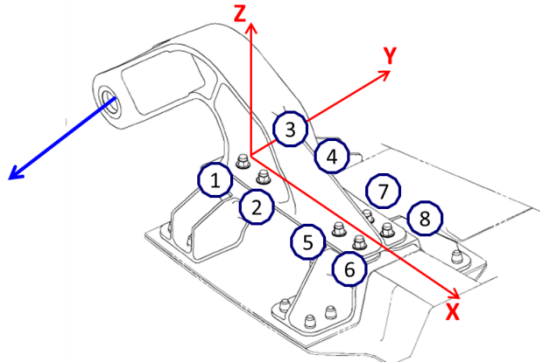
All fasteners are equal, so area (A) is the same for all of them

$$x_c = \frac{\sum x_i \cdot A_i}{\sum A_i} = \frac{0 + 32 + 0 + 32 + 150 + 182 + 150 + 182}{8} = 91 \text{ mm}$$
$$y_c = \frac{\sum y_i \cdot A_i}{\sum A_i} = \frac{-16 - 16 + 16 + 16 - 16 - 16 + 16 + 16}{8} = 0 \text{ mm}$$

X (mm)	Y (mm)	Z (mm)
91	0	0

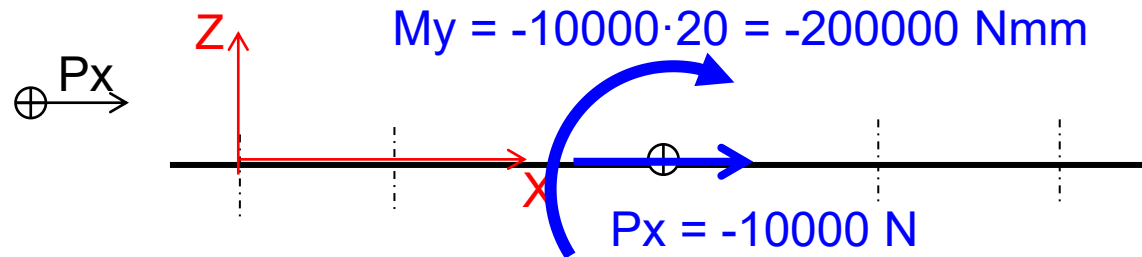
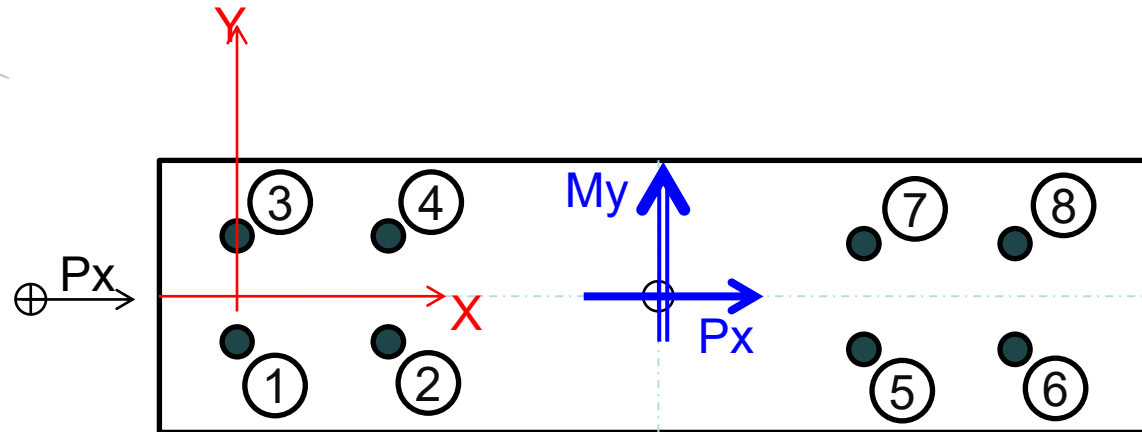
Bolt Group Example

Forces and moments in the centroid



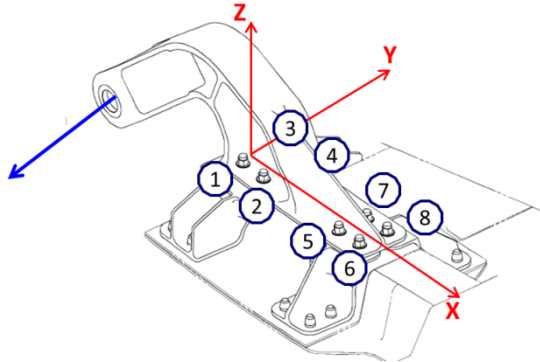
Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

Load Case	Px (N)	Py (N)	Pz (N)
Tx	-10000	0	0
Lug location: (-150,0,20) mm			



Bolt Group Example

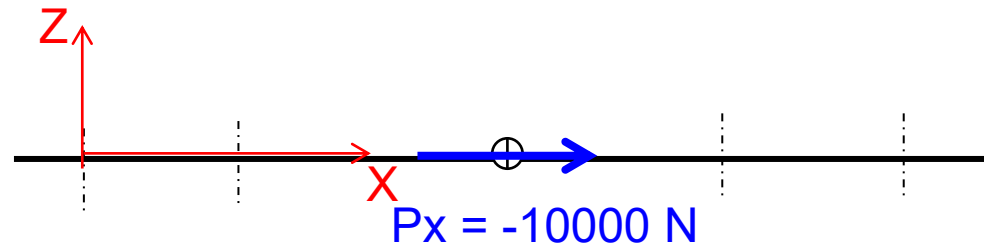
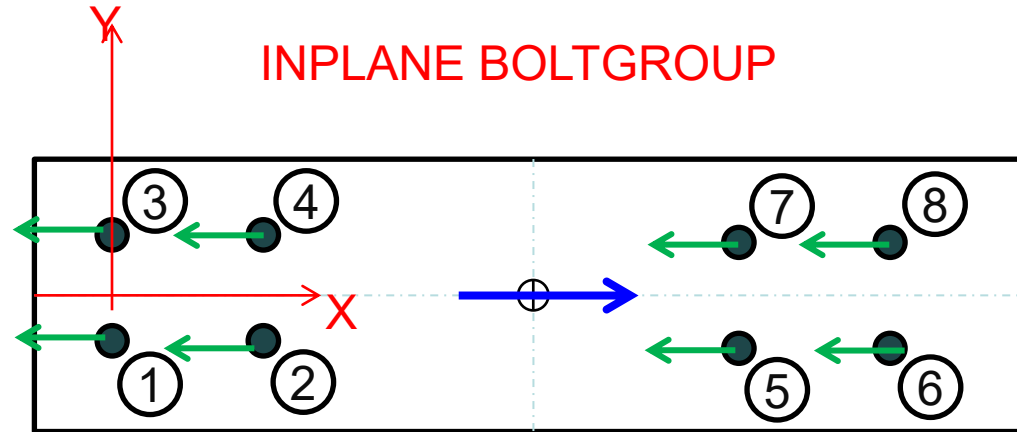
Equilibrium of forces



Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

Load Case	Px (N)	Py (N)	Pz (N)
Tx	-10000	0	0
Lug location: (-150,0,20) mm			

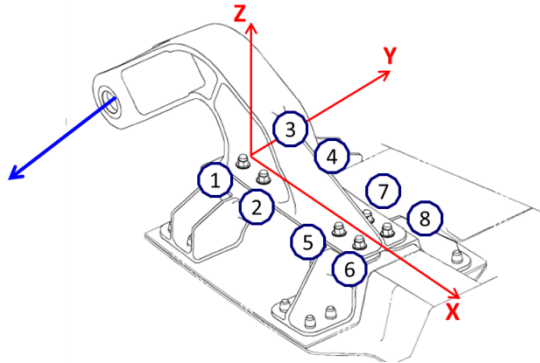
INPLANE BOLTGROUP



$$P_{x_{fast}} = \frac{P_x}{n^o_{fast}} = \frac{-10000}{8} = -1250N$$

Bolt Group Example

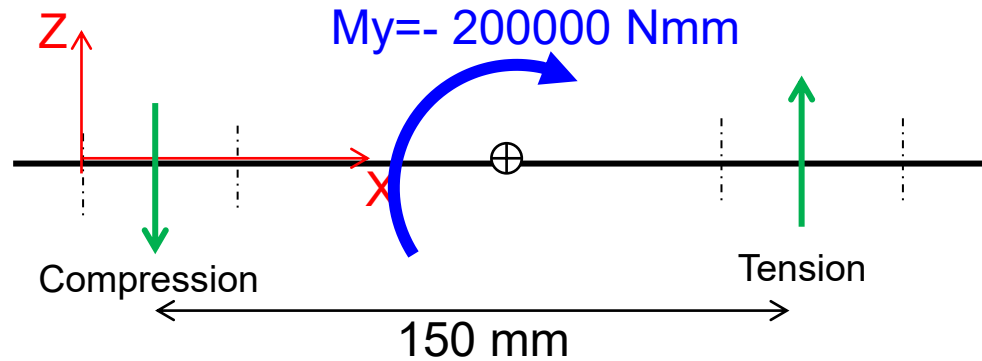
Equilibrium of moments



Load Case	Px (N)	Py (N)	Pz (N)
Tx	-10000	0	0
Lug location: (-150,0,20) mm			

Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

OFFPLANE BOLTGROUP

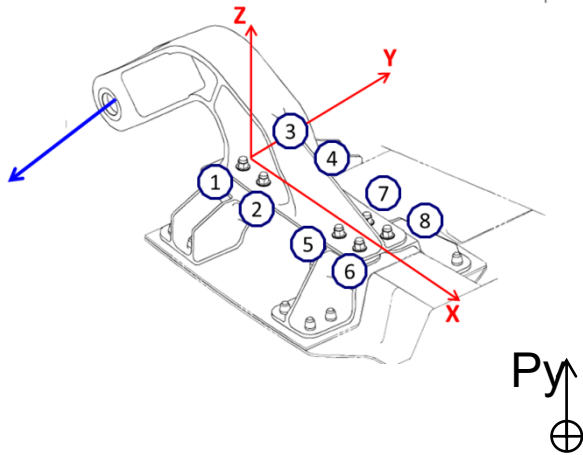


$$P_{tension} = \frac{M_y}{\text{distance between fasteners group}} = \frac{200000}{150} = 1333N$$

$$P_{tn_{fast}} = \frac{P_{tension}}{n^{\circ}bolts \text{ in tension}} = \frac{1333}{4} = 333,3N$$

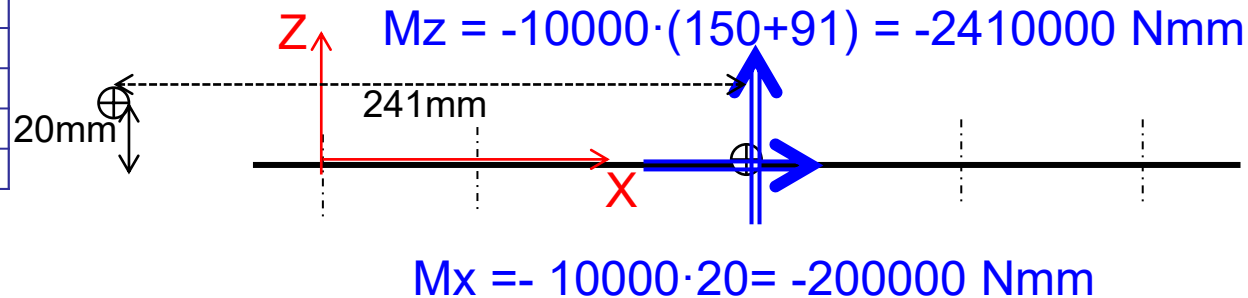
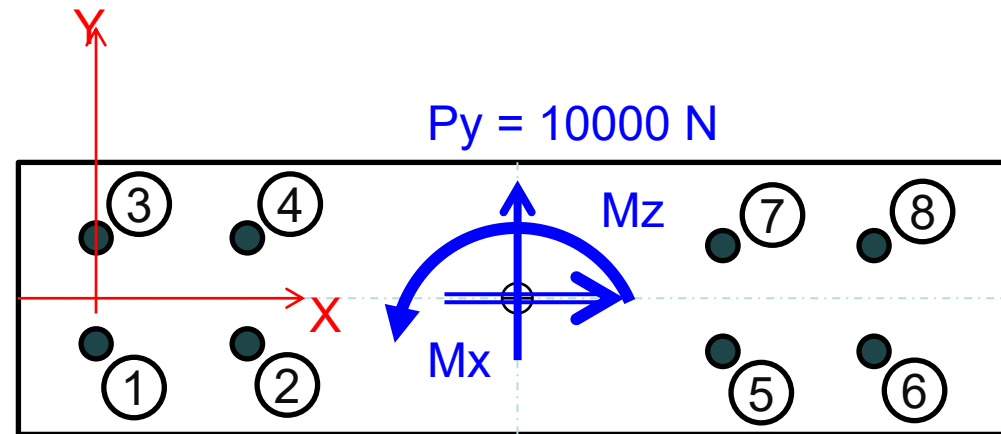
Bolt Group Example

Forces and moments in the centroid



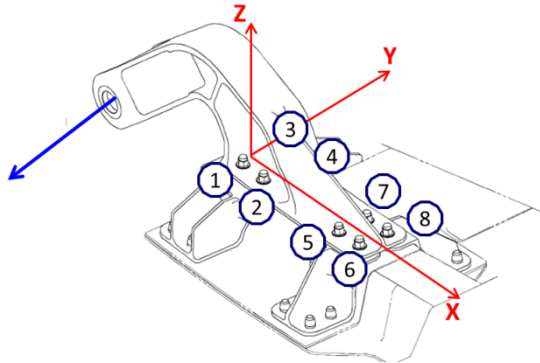
Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

Load Case	Px (N)	Py (N)	Pz (N)
Ty	0	10000	0
Lug location: (-150,0,20) mm			



Bolt Group Example

Equilibrium of forces

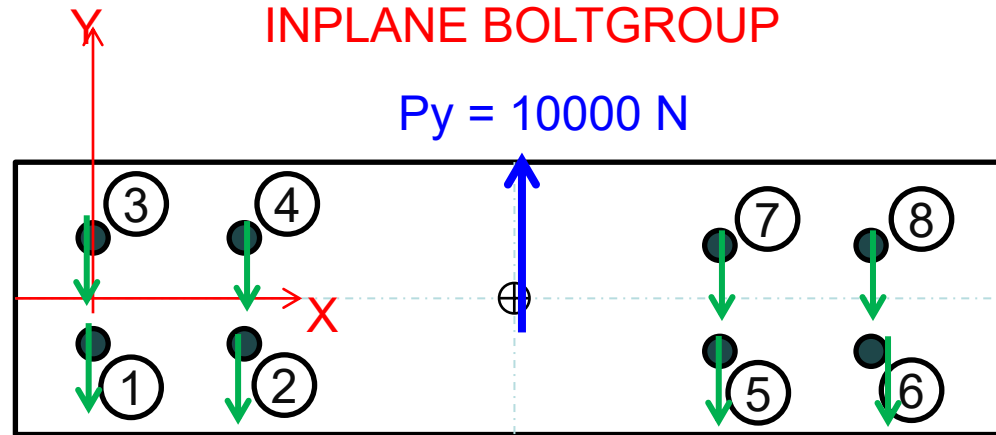


Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

Load Case	Px (N)	Py (N)	Pz (N)
Ty	0	10000	0
Lug location: (-150,0,20) mm			

INPLANE BOLTGROUP

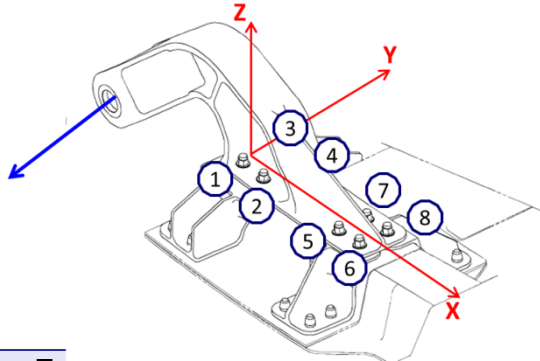
$P_y = 10000 \text{ N}$



$$P_{y_{fast}} = \frac{P_y}{n^{\circ}fast} = \frac{10000}{8} = 1250N$$

Bolt Group Example

Equilibrium of moments (Mz)

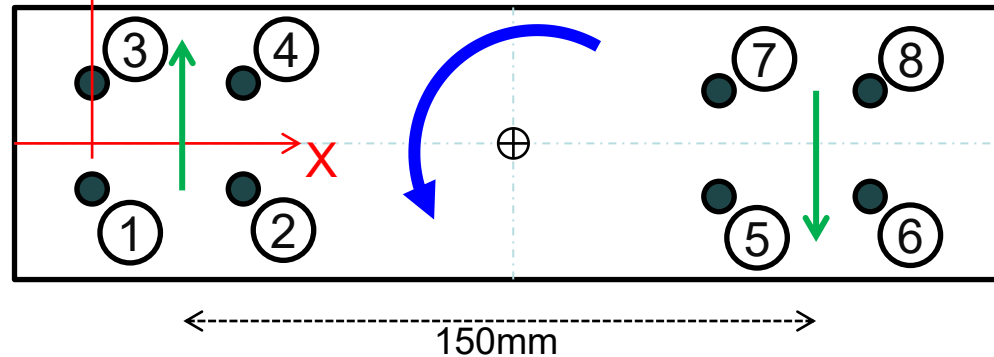


Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

Load Case	Px (N)	Py (N)	Pz (N)
Ty	0	10000	0
Lug location: (-150,0,20) mm			

INPLANE BOLTGROUP

Mz = -2410000 Nmm

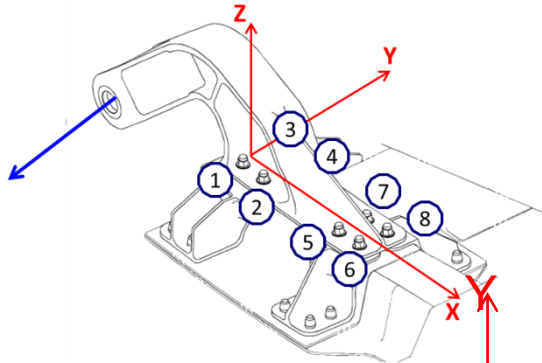


$$P_{Y,fast1,2,3,4} = -P_{Y,fast5,6,7,8} = \frac{M_z}{\text{distance between fasteners group}} = \frac{-2410000}{150} = -16067N$$

$$P_{y,fast} = \frac{P_Y}{n^{\circ}bolts \text{ per group}} = \pm \frac{16067}{4} = 4017N$$

Bolt Group Example

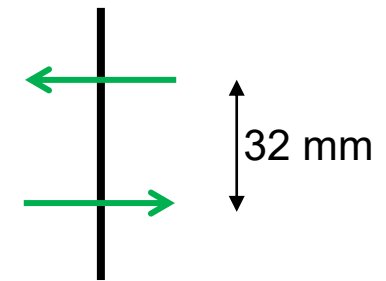
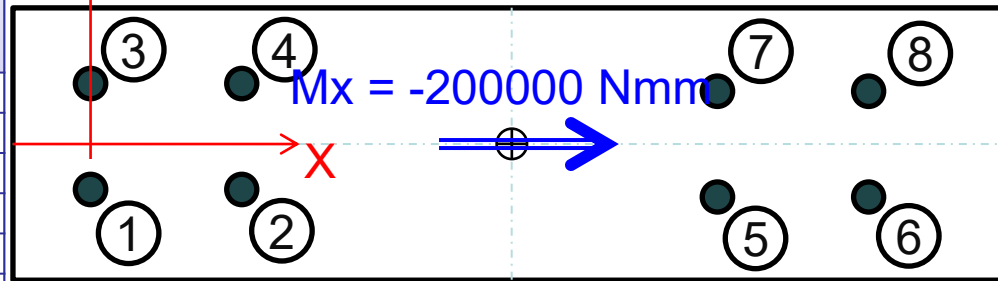
Forces and moments (Mx)



Load Case	Px (N)	Py (N)	Pz (N)
Ty	0	10000	0
Lug location: (-150,0,20) mm			

OFFPLANE BOLTGROUP

Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

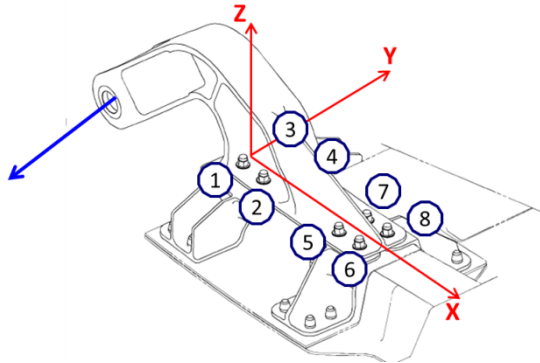


$$P_{Z,fast1,2,5,6} = -P_{Z,fast3,4,7,8} = \frac{M_x}{\text{distance between fasteners group}} = \frac{200000}{32} = 6250N$$

$$P_{Z,fast} = \frac{P_z}{n^{\circ}\text{bolts per group}} = \pm \frac{6250}{4} = \pm 1563N$$

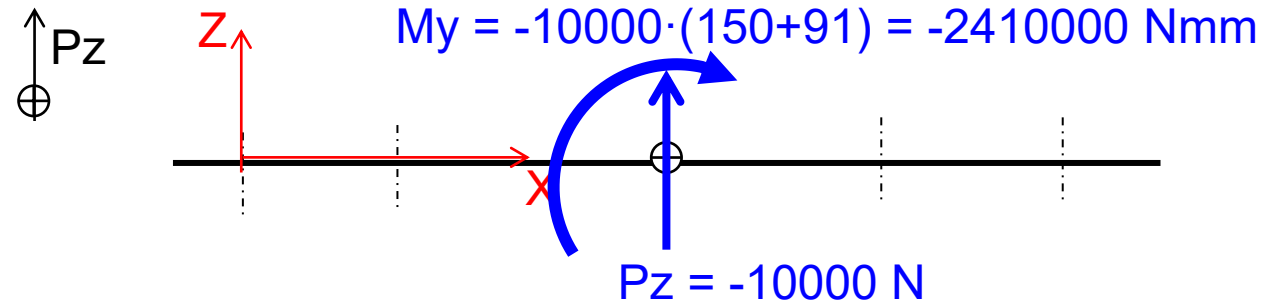
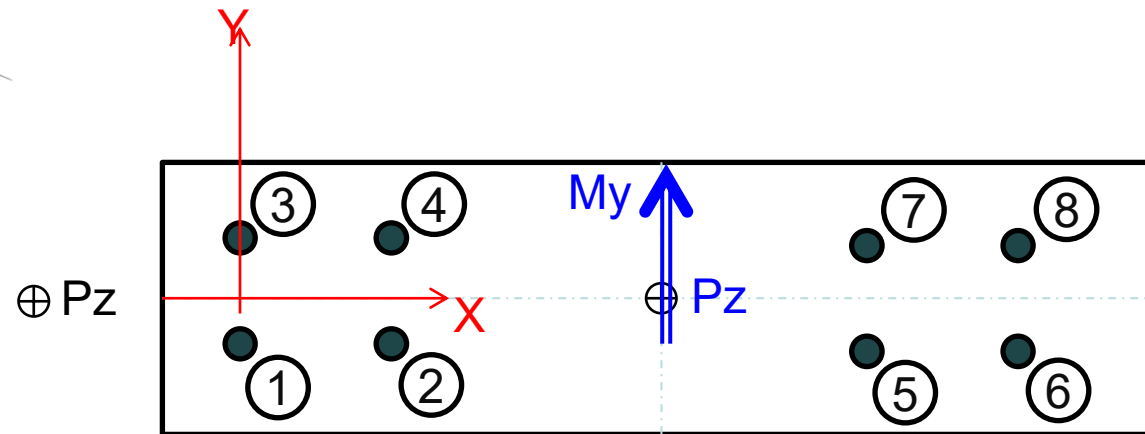
Bolt Group Example

Forces and moments in the centroid



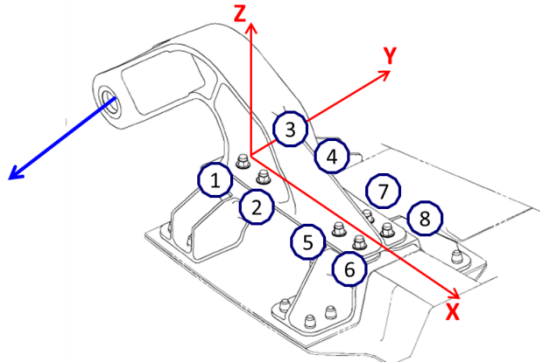
Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

Load Case	Px (N)	Py (N)	Pz (N)
Tz	0	0	-10000
Lug location: (-150,0,20) mm			



Bolt Group Example

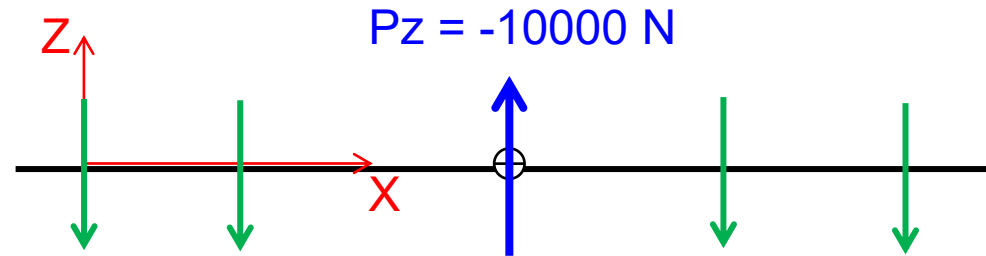
Equilibrium of forces



Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

Load Case	Px (N)	Py (N)	Pz (N)
Tz	0	0	-10000
Lug location: (-150,0,20) mm			

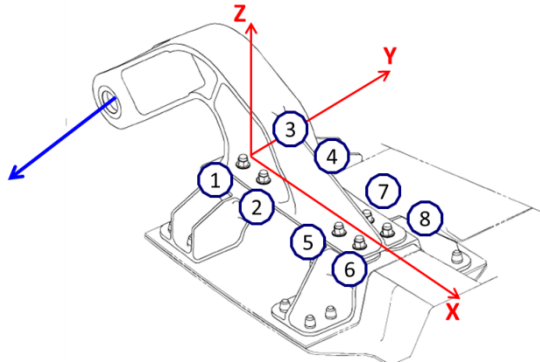
OFFPLANE BOLTGROUP



$$P_{z_{fast}} = \frac{P_z}{n^{\circ}fast} = \frac{-10000}{8} = -1250N$$

Bolt Group Example

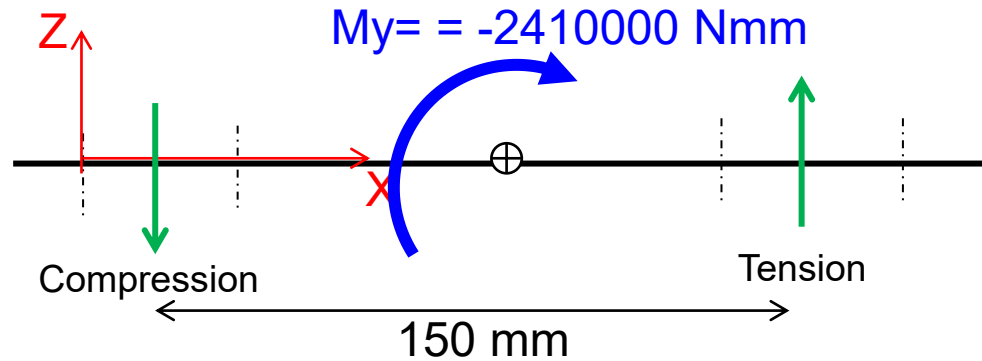
Equilibrium of moments



Load Case	Px (N)	Py (N)	Pz (N)
Tz	0	0	-10000
Lug location: (-150,0,20) mm			

Fastener location			
Fast	X (mm)	Y (mm)	Z (mm)
1	0	-16	0
2	32	-16	0
3	0	16	0
4	32	16	0
5	150	-16	0
6	182	-16	0
7	150	16	0
8	182	16	0

OFFPLANE BOLTGROUP



$$P_{tension} = \frac{M_y}{\text{distance between fasteners group}} = \frac{2410000}{150} = 16067N$$

$$P_{tn_{fast}} = \frac{P_{tension}}{n^{\circ}bolts \text{ in tension}} = \frac{16067}{4} = 4017N$$

Bolt Group Example

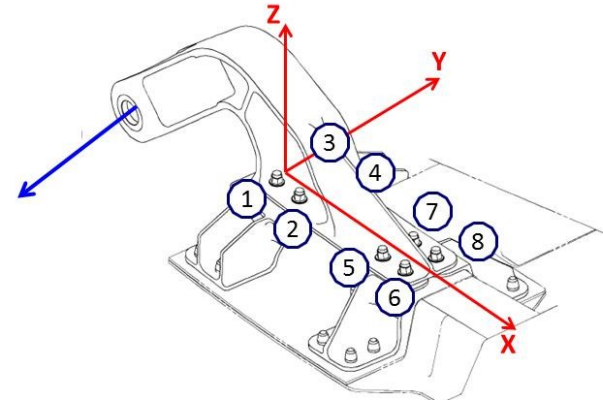
2. Bolt group moment of inertia calculation:

$$I_x = \sum (Y_i - Y_{CR})^2 \cdot A_i ; I_y = \sum (X_i - X_{CR})^2 \cdot A_i ; I_p = I_x + I_y$$

Where,

X_i ; Y_i (mm): Bolt coordinates

X_{CR} ; Y_{CR} (mm): Center of riveting coordinates



Bolts	$(X_i - X_{CR})$ (mm)	$(Y_i - Y_{CR})$ (mm)	$(X_i - X_{CR})^2$ (mm ²)	$(Y_i - Y_{CR})^2$ (mm ²)
1	-91	-16	8281	256
2	-59	-16	3481	256
3	-91	16	8281	256
4	-59	16	3481	256
5	59	-16	3481	256
6	91	-16	8281	256
7	59	16	3481	256
8	91	16	8281	256

I_x (mm ⁴)	2048
I_y (mm ⁴)	47048
I_p (mm ⁴)	49096

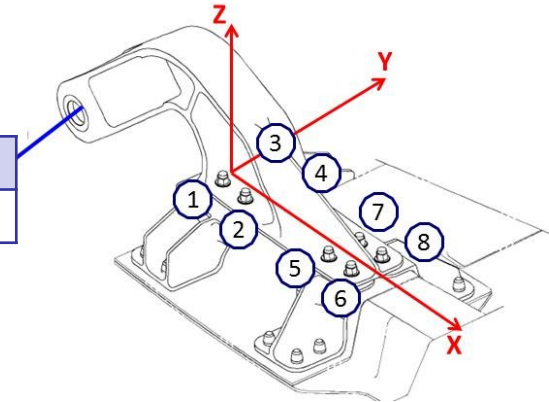
3th SOLUTION: EXACT

Bolt Group Example

3. Forces and moments in the centroid:

Applied load with origin in the centroid

X (mm)	Y (mm)	Z (mm)
-150	0	20



Forces and moments in the centroid:

Load Case	Px (N)	Py (N)	Pz (N)	Mx (Nmm)	My (Nmm)	Mz (Nmm)
Tx	-10000	0	0	0	-200000	0
Ty	0	10000	0	-200000	0	-2410000
Tz	0	0	-10000	0	-2410000	0

Bolt Group Example

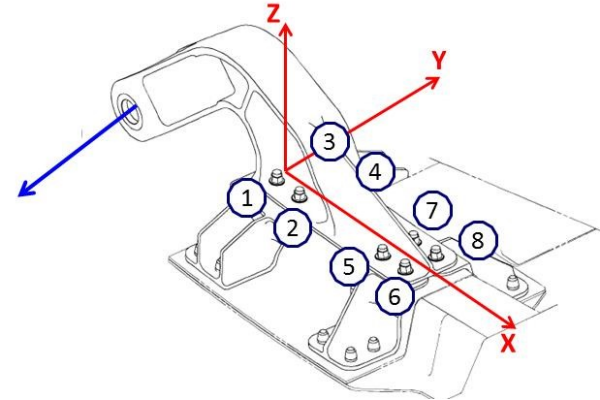
4. Axial bolt loads: OFFPLANE BOLTGROUP

Due to axial force: $P_{z_i}^{F_{z_CR}} = \frac{F_{z_CR} \cdot K_{ti}}{\sum K_{ti}}$

Due to bending (Mx): $P_{z_Mx} = \frac{M_{x_CR} \cdot K_{ti} \cdot (Y_i - Y_{CR})}{I_x}$

Due to bending (My): $P_{z_My} = - \frac{M_{y_CR} \cdot K_{ti} \cdot (X_i - X_{CR})}{I_y}$

With Kti equal for all bolts



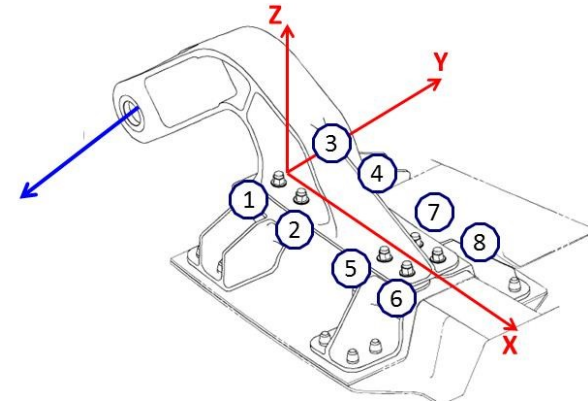
LC	Tx				Ty				Tz			
Bolts	Pz (N), due to			Total	Pz (N), due to			Total	Pz (N), due to			Total
	Pz	Mx	My	Pz (N)	Pz	Mx	My	Pz (N)	Pz	Mx	My	Pz (N)
1	0	0	-387	-387	0	1563	0	1563	-1250	0	-4661	-5911
2	0	0	-251	-251	0	1563	0	1563	-1250	0	-3022	-4272
3	0	0	-387	-387	0	-1563	0	-1563	-1250	0	-4661	-5911
4	0	0	-251	-251	0	-1563	0	-1563	-1250	0	-3022	-4272
5	0	0	251	251	0	1563	0	1563	-1250	0	3022	1772
6	0	0	387	387	0	1563	0	1563	-1250	0	4661	3411
7	0	0	251	251	0	-1563	0	-1563	-1250	0	3022	1772
8	0	0	387	387	0	-1563	0	-1563	-1250	0	4661	3411

5. X-loads:

INPLANE BOLTGROUP

$$\text{Due to X force: } P_{x_i}^{F_{x_CR}} = \frac{F_{x_CR} \cdot A_i}{\sum A_i}$$

$$\text{Due to in-plane moment (Mz): } P_{x_i}^{M_{z_CR}} = - \frac{M_{z_CR} \cdot A_i \cdot (Y_i - Y_{CR})}{I_p}$$



With same Area for all bolts

Load case	LC: Tx			LC: Ty			LC: Tz		
Bolts	Px (N), due to		Total	Px (N), due to		Total	Px (N), due to		Total
	Px	Mz	Px (N)	Px	Mz	Px (N)	Px	Mz	Px (N)
1	-1250	0	-1250	0	-785	-785	0	0	0
2	-1250	0	-1250	0	-785	-785	0	0	0
3	-1250	0	-1250	0	785	785	0	0	0
4	-1250	0	-1250	0	785	785	0	0	0
5	-1250	0	-1250	0	-785	-785	0	0	0
6	-1250	0	-1250	0	-785	-785	0	0	0
7	-1250	0	-1250	0	785	785	0	0	0
8	-1250	0	-1250	0	785	785	0	0	0

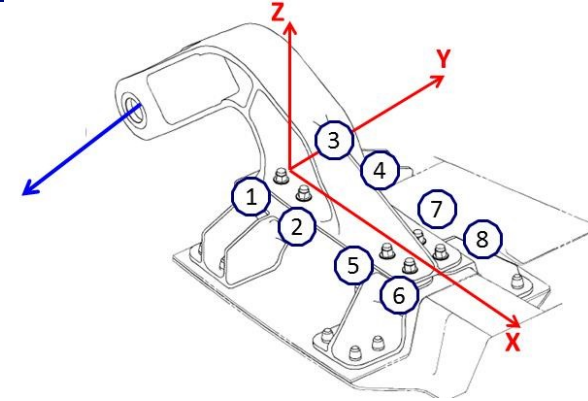
Bolt Group Example

6. Y-loads:

INPLANE BOLTGROUP

$$\text{Due to Y force: } P_{y_i}^{F_{y_CR}} = \frac{F_{y_CR} \cdot A_i}{\sum A_i}$$

$$\text{Due to in-plane moment (Mz): } P_{y_i}^{M_{z_CR}} = \frac{M_{z_CR} \cdot A_i \cdot (X_i - X_{CR})}{I_p}$$



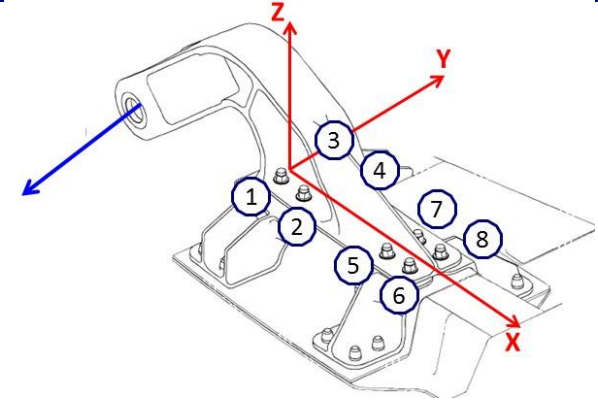
With same Area for all bolts

Load case	LC: Tx			LC: Ty			LC: Tz		
Bolts	Py (N), due to		Total	Py (N), due to		Total	Py (N), due to		Total
	Py	Mz	Py (N)	Py	Mz	Py (N)	Py	Mz	Py (N)
1	0	0	0	1250	4467	5717	0	0	0
2	0	0	0	1250	2896	4146	0	0	0
3	0	0	0	1250	4467	5717	0	0	0
4	0	0	0	1250	2896	4146	0	0	0
5	0	0	0	1250	-2896	-1646	0	0	0
6	0	0	0	1250	-4467	-3217	0	0	0
7	0	0	0	1250	-2896	-1646	0	0	0
8	0	0	0	1250	-4467	-3217	0	0	0

Bolt Group Example

7. Bolt load:

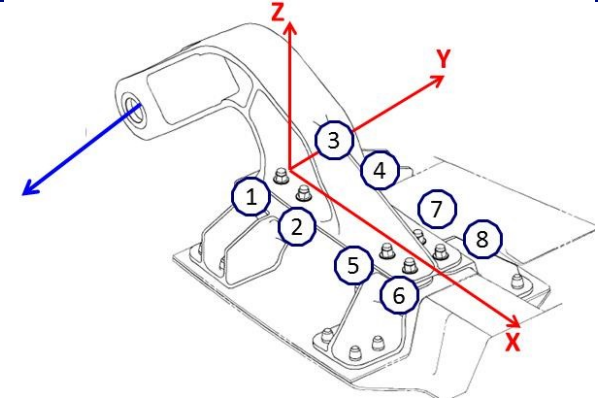
Load Case	Px (N)	Py (N)	Pz (N)
Tx	-10000	0	0
Ty	0	10000	0
Tz	0	0	-10000
Lug location: (-150,0,20) mm			



Bolts	LC: Tx			LC: Ty			LC: Tz		
	Px (N)	Py (N)	Pz (N)	Px (N)	Py (N)	Pz (N)	Px (N)	Py (N)	Pz (N)
1	-1250	0	-387	-785	5717	1563	0	0	-5911
2	-1250	0	-251	-785	4146	1563	0	0	-4272
3	-1250	0	-387	785	5717	-1563	0	0	-5911
4	-1250	0	-251	785	4146	-1563	0	0	-4272
5	-1250	0	251	-785	-1646	1563	0	0	1772
6	-1250	0	387	-785	-3217	1563	0	0	3411
7	-1250	0	251	785	-1646	-1563	0	0	1772
8	-1250	0	387	785	-3217	-1563	0	0	3411

Attachment Analysis

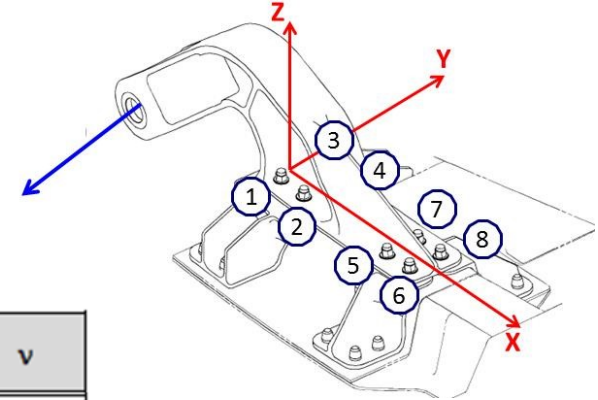
Bolts	LC: Ty		
	Px (N)	Py (N)	Pz (N)
1	-785	5717	1563
2	-785	4146	1563
3	785	5717	-1563
4	785	4146	-1563
5	-785	-1646	1563
6	-785	-3217	1563
7	785	-1646	-1563
8	785	-3217	-1563



Maximum shear load $P_{sh,1} = \sqrt{P_x^2 + P_y^2} = 5770N$, with $P_{tn} = 1563N$

Attachment Analysis

Fitting material: Aluminum (**Exercise 4**)



F_{tu} (MPa)	F_{ty} (MPa)	F_{cy} (MPa)	F_{su} (MPa)	F_{bru} (MPa)	E_t (MPa)	E_c (MPa)	e%	ν
468.8	386.1	365.4	282.7	903.2	71016	73084	10	0.33

Fastener properties:

Diameter code	Diameter mm	Double Shear strength (N)	Tensile strength (N)
			Protruding head
4	6.4	41330	25900

Nut properties:

Diameter code	Diameter mm	Nut tensile strength allowable (N)
4	6.4	22250

$$0.5 < t/D < 2.0, \text{ so } 0.5 \cdot 6,4 < t < 2 \cdot 6,4; 3,2 \text{ mm} < t < 12,8 \text{ mm}$$

Bearing analysis:

$$P_{br,all} = F_{bru} (e/D=2,0) \cdot D \cdot t_{base} > FF \cdot P_{sh} = 5770 \text{ N}$$

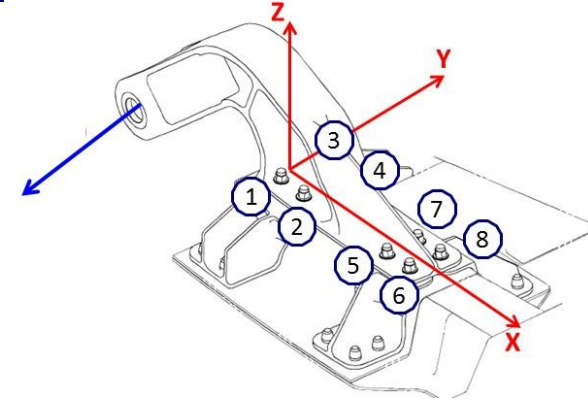
$$\text{So, } t_{base} \geq \frac{5770 \cdot 1,15}{903,2 \cdot 6,35} = 1,2 \text{ mm}$$

$$\text{for } t=3,2 \text{ mm } RF = \frac{903,2 \cdot 6,35 \cdot 3,2}{1,15 \cdot 5770} = \frac{18353}{1,15 \cdot 5770} = 2,76$$

Bolt analysis, shear:

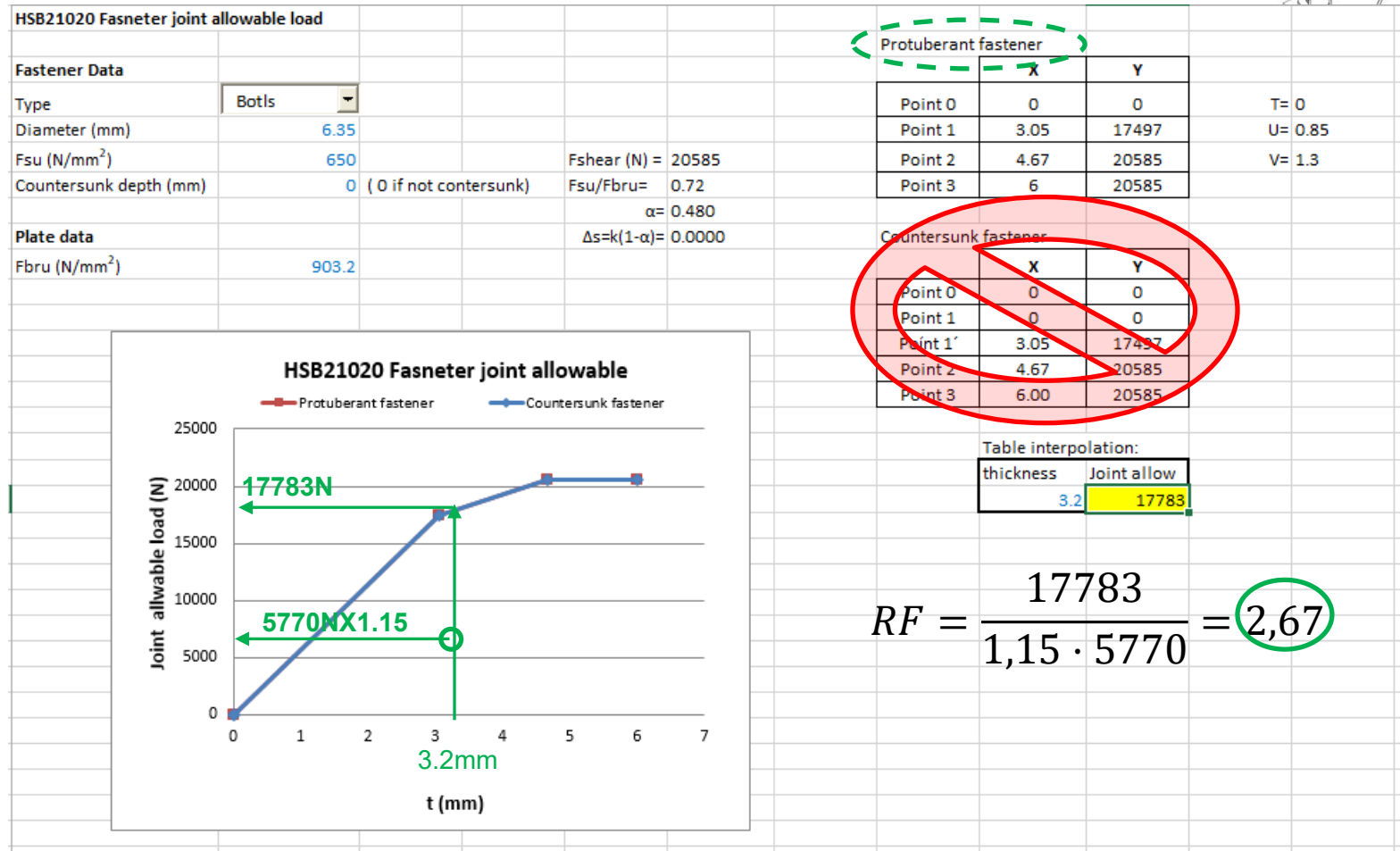
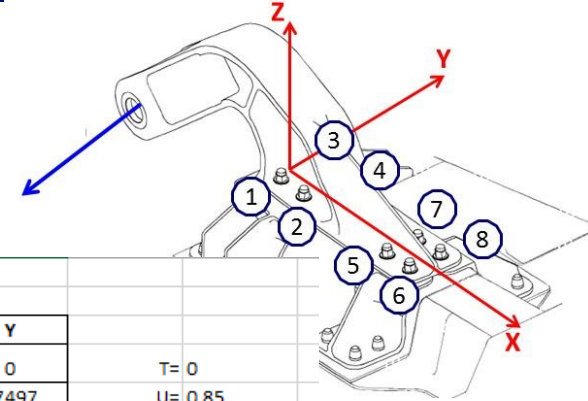
$$P_{su} = 41330/2 = 20665 \text{ N}$$

$$RF = \frac{20665}{1,15 \cdot 5770} = 3,11$$



Attachment Analysis

Bolt bearing combination:

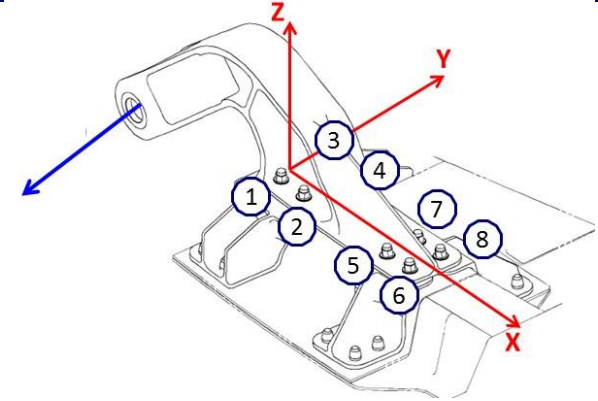


Attachment Analysis

Bolt analysis, tension

$$P_{tu} = 25900 \text{ N}$$

$$RF = \frac{25900}{1,15 \cdot 1563} = 14,40 \gg 2,00$$

Nut analysis, tension

$$P_{tu} = 22250 \text{ N}$$

$$RF = \frac{22250}{1,15 \cdot 1563} = 12,37 \gg 2,00$$

$$\text{Maximum shear load } P_{sh,1} = \sqrt{P_x^2 + P_y^2} = 5770 \text{ N}, \text{ with } P_{tn} = 1563 \text{ N}$$

These are all RF for LC2 (Ty) but there is a worst tensión LC with +3411N (LC3, Tz)

Attachment Analysis

Bolt analysis, tension+shear combination

$$P_{tu} = 25900 \text{ N}$$

$$P_{su} = 20665 \text{ N}$$

$$RF = \frac{1}{1,15 \sqrt{\left(\frac{1563}{25900}\right)^2 + \left(\frac{5770}{20665}\right)^2}} = 3,04$$

Pull-out analysis:

$$P_{pull} = \frac{P}{0,6 \cdot \pi \cdot D_w \cdot t} = \frac{1563}{0,6 \cdot \pi \cdot 12,7 \cdot t} = \frac{65,3}{t}$$

$$RF = \frac{F_{su}}{FF \cdot P_{pull}} = \frac{282,7}{1,15 \cdot 65,3} \cdot t > 1,00$$

$$\text{So, } t > 0,3 \text{ mm}$$

